

Gebze Technical University

Department of Electronics Engineering

ELM367: Digital Signal Processing

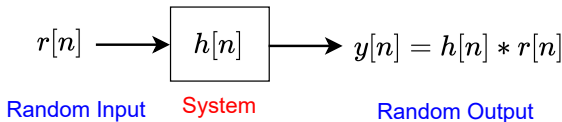
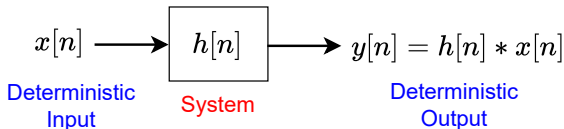
Lecture 6: Discrete-Time Random Signals

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Discrete-Time Random Signals

Introduction



- ▶ Deterministic Signals: Mathematical Expressions
- ▶ Random Signals: No clear Mathematical Expressions
 - ▶ We can rely on the statistical characteristics to process the signal

Discrete-Time Random Signals

Introduction

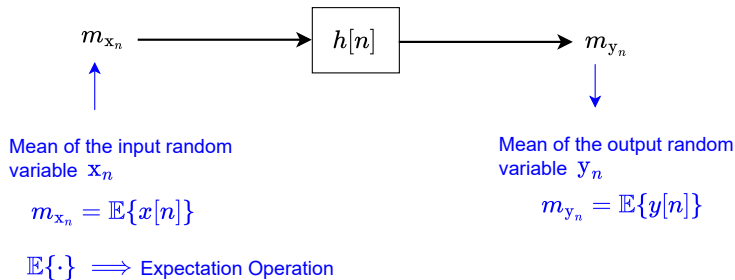
Specifically, the following statistical parameters can be useful for random signal processing:

- ▶ Expected value (Mean)
- ▶ Variance
- ▶ Probability density function (PDF)
- ▶ Joint PDF
- ▶ Auto-correlation
- ▶ Power spectral density (PSD)

We will assume the Wide-sense stationary random process, thus, we mainly focus on [the Mean](#), and [Autocorrelation function](#)

Discrete-Time Random Signals

Relationship between Means of the Input and Output Random signals



For a wide-sense stationary random process:

- ▶ m_{x_n} and m_{y_n} are independent of time n , therefore they can be simply written as m_x and m_y , respectively.

Discrete-Time Random Signals

Relationship between Means of the Input and Output Random signals

Recall the convolution sum formula for an LTI system:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$\mathbb{E}\{y[n]\} = \mathbb{E}\left\{\sum_{k=-\infty}^{\infty} h[k]x[n-k]\right\}$$

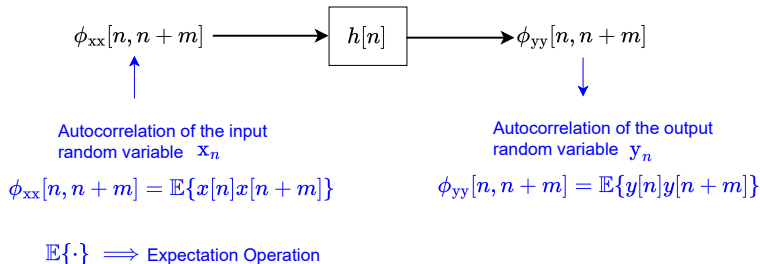
$$\mathbb{E}\{y[n]\} = \sum_{k=-\infty}^{\infty} \mathbb{E}\{h[k]x[n-k]\}$$

$$\mathbb{E}\{y[n]\} = \sum_{k=-\infty}^{\infty} h[k]\mathbb{E}\{x[n-k]\}$$

$$m_y = m_x \sum_{k=-\infty}^{\infty} h[k] = m_x \sum_{k=-\infty}^{\infty} h[k] \cdot \underbrace{e^{-j0k}}_{\omega=0} = H(e^{j0})m_x$$

Discrete-Time Random Signals

Relationship between Autocorrelations of the Input and Output Random signals



For a wide-sense stationary random process:

- ▶ $\phi_{xx}[n, n+m] = \phi_{xx}[m]$ i.e., Autocorrelation of a signal at different times is a function of the time difference only

Discrete-Time Random Signals

Relationship between Autocorrelations of the Input and Output Random signals

Let's analyze the Autocorrelation of the output random signal:

$$\begin{aligned}\phi_{yy}[n, n + m] &= \mathbb{E} \{y[n] \cdot y[n + m]\} \\&= \mathbb{E} \left\{ \sum_{k=-\infty}^{\infty} h[k]x[n - k] \cdot \sum_{r=-\infty}^{\infty} h[r]x[n + m - r] \right\} \\&= \mathbb{E} \left\{ \sum_{k=-\infty}^{\infty} h[k] \sum_{r=-\infty}^{\infty} h[r]x[n - k]x[n + m - r] \right\} \\&= \sum_{k=-\infty}^{\infty} h[k] \sum_{r=-\infty}^{\infty} h[r] \underbrace{\mathbb{E} \{x[n - k]x[n + m - r]\}}_{=\phi_{xx}[m-r+k]} \\&= \sum_{k=-\infty}^{\infty} h[k] \sum_{r=-\infty}^{\infty} \phi_{xx}[m - r + k]h[r]\end{aligned}$$

Discrete-Time Random Signals

Relationship between Autocorrelations of the Input and Output Random signals

Let $\ell = r - k$

$$\begin{aligned}\phi_{yy}[m] &= \sum_{\ell=-\infty}^{\infty} \phi_{xx}[m - \ell] \underbrace{\sum_{k=-\infty}^{\infty} h[k]h[\ell + k]}_{C_{hh}[\ell]} \\ &= \underbrace{\sum_{\ell=-\infty}^{\infty} \phi_{xx}[m - \ell] C_{hh}[\ell]}_{\text{Convolution Sum}} = C_{hh}[m] * \phi_{xx}[m]\end{aligned}$$

where

$$C_{hh}[\ell] = \sum_{k=-\infty}^{\infty} h[k]h[\ell + k] = \sum_{k=-\infty}^{\infty} h[k]h[\ell - (-k)] = h[\ell] * h[-\ell]$$

$C_{hh}[\ell]$ is referred to as **Deterministic autocorrelation** or **Aperiodic autocorrelation** of $h[\ell]$

Discrete-Time Random Signals

Relationship between Autocorrelations of the Input and Output Random signals

In DTFT Domain

$$\begin{aligned}\phi_{yy}[m] &= C_{hh}[m] * \phi_{xx}[m] \\ &\quad \downarrow \mathcal{F} \\ \underbrace{\Phi_{yy}(e^{j\omega})}_{\text{Power Spectral density}} &= C_{hh}(e^{j\omega})\Phi_{xx}(e^{j\omega})\end{aligned}$$

from

$$\begin{aligned}C_{hh}[n] &= h[n] * h[-n] \\ &\quad \downarrow \mathcal{F} \\ C_{hh}(e^{j\omega}) &= H(e^{j\omega})H^*(e^{j\omega}) = |H(e^{j\omega})|^2\end{aligned}$$

Therefore:

$$\Phi_{yy}(e^{j\omega}) = |H(e^{j\omega})|^2\Phi_{xx}(e^{j\omega})$$

Discrete-Time Random Signals

Average Power

For random signals, we are more interested in the average power, i.e. $\mathbb{E}\{y^2[n]\}$:

$$\begin{aligned}\mathbb{E}\{y^2[n]\} &= \mathbb{E}\{y[n]y[n]\} \\ &= \phi_{yy}[0]\end{aligned}$$

Using the DTFT relationship:

$$\begin{aligned}&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi_{yy}(e^{j\omega}) e^{j\omega n} d\omega \Big|_{n=0} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi_{yy}(e^{j\omega}) d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 \Phi_{xx}(e^{j\omega}) d\omega\end{aligned}$$

Discrete-Time Random Signals

Cross Correlation Between Input and Output Signals

$$\begin{aligned}\phi_{xy}[n, n+m] &= \mathbb{E} \{x[n] \cdot y[n+m]\} \\&= \mathbb{E} \left\{ x[n] \sum_{k=-\infty}^{\infty} h[k] x[n+m-k] \right\} \\&= \sum_{k=-\infty}^{\infty} h[k] \mathbb{E} \{x[n] x[n+m-k]\} \\&= \sum_{k=-\infty}^{\infty} h[k] \phi_{xx}[m-k] = h[n] * \phi_{xx}[m]\end{aligned}$$

If we take Fourier transform

$$\Phi_{xy}(e^{j\omega}) = H(e^{j\omega}) \Phi_{xx}(e^{j\omega})$$

Discrete-Time Random Signals

Examples

Example-1: Given a random input signal with autocorrelation $\phi_{xx}[m] = \sigma_x^2 \delta[m]$. Find:

1. Power spectral density $\Phi_{xx}(e^{j\omega})$ of this signal
2. Average power of this input signal
3. Given $h[n] = a^n u[n]$, find power spectral density of the output signal, i.e., $\Phi_{yy}(e^{j\omega})$

Discrete-Time Random Signals

Examples

Example-2: Given a wide-sense stationary discrete-time random signal $x[n]$ with mean 0.5 and autocorrelation function given as

$$\phi_{xx}[n] = \begin{cases} 1 & n = 0 \\ \frac{1}{5} & n = \pm 1 \\ 0 & \text{Otherwise} \end{cases}$$

This signal is applied to the LTI system defined by a difference equation $y[n] = 0.5x[n] + 0.5x[n - 1]$

1. Find Impulse response of the system
2. Find the mean of the output signal $y[n]$
3. Find autocorrelation of the output signal
4. Find Power spectrum density of the output signal

Discrete-Time Random Signals

Examples

Example-3: A discrete-time random process $x[n]$ is generated from a sequence of uniformly distributed random numbers generator from interval of $[-2,2]$. The discrete random samples $x[n]$ has autocorrelation of $\phi_{xx}[m] = \frac{3}{4}\delta[m]$ and through an LTI system to get $y[n]$. The impulse response of the system is given by $h[n] = \delta[n] - \delta[n - 1]$. Determine:

1. Mean of the discrete output process
2. power spectral density (PSD) of the output $\Phi_{yy}(e^{j\omega})$
3. Autocorrelation of the output process $\phi_{yy}[n]$